

Module 1 – Generative AI Essentials

Demo 1 - Creating a Small Transformer using PyTorch

Problem Statement

Scenario:

RetailVerse, a growing e-commerce platform, receives thousands of customer reviews each month across its diverse product catalog. The company's support and product teams aim to quickly assess customer sentiment to prioritize feature improvements, resolve complaints efficiently, and highlight positive trends. However, the existing process of manually labeling sentiment in reviews is time-consuming, inconsistent, and not scalable with the company's growth.

To address this challenge, RetailVerse's machine learning team has been assigned the task of developing a lightweight, transparent, and self-contained sentiment analysis model that does not depend on large-scale external APIs. As an initial step, the team chooses to implement a **mini transformer model using PyTorch**, trained on internally curated customer review data.



The objective of this initiative is to classify each review into one of three sentiment categories—**positive**, **neutral**, or **negative**, using a compact transformer architecture built from scratch and trained on a controlled dataset.

Use the sample dataset (sentiment_dataset.csv) provided to support model training and evaluation for this problem statement.